

**BANARAS HINDU UNIVERSITY**

B.Sc. (Hons.) Semester IV Examination, 2023-24

**Paper No. CHB-401: Inorganic & Organic Chemistry-II**

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer five questions in total, including Question 1 from each section which is compulsory. Use separate answer books for Section A and Section B.*

**SECTION A**

(Inorganic Chemistry-II – Marks: 35)

**Question 1.** Answer all parts of the following: [5 × 2 = 10]

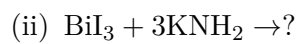
- (a) Write all oxidation states shown by the lanthanides Tb and Yb with proper reasoning.
- (b) Arrange HBr, HNO<sub>3</sub>, and HClO<sub>4</sub> according to their acidic strength in: (i) water and (ii) acetic acid. Justify.
- (c) Why is anhydrous CuSO<sub>4</sub> colorless and CuSO<sub>4</sub> · 5H<sub>2</sub>O blue? Explain.
- (d) Solvents having high dielectric constants are suitable for dissolving ionic solids. Why?
- (e) Give two examples of hard and soft acids according to the HSAB principle.

**Question 2.**

- (a) Why do 3d transition metals show variable oxidation states? How can their low and high oxidation states be stabilized? Give examples. [6]
- (b) State the characteristic features of dilute solutions of sodium in liquid ammonia. How does the electrical conductivity change with increasing concentration of sodium? [6]

**Question 3.**

- (a) Write the IUPAC names of the following coordination complexes: [4]
  - (i) Na<sub>3</sub>[Co(NO<sub>2</sub>)<sub>6</sub>]
  - (ii) [Cr(en)<sub>3</sub>]Cl<sub>3</sub>
  - (iii) K[PtCl<sub>3</sub>(C<sub>2</sub>H<sub>4</sub>)]
- (b) Draw structures of all possible geometrical and optical isomers of [Co(en)<sub>2</sub>Cl<sub>2</sub>]<sup>+</sup>. [4]
- (c) Write the products of the following reactions in liquid ammonia: [4]
  - (i) AgCl + KNH<sub>2</sub> →?



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**Question 4.**

- (a)  $\text{AsF}_5$  behaves as an acid in  $\text{BrF}_3$  solvent. Justify this with chemical equations. [6]
- (b) Why is the separation of lanthanides difficult? Describe the ion-exchange method for the separation of lanthanide ions. [6]

**SECTION B**

(Organic Chemistry-III – Marks: 35)

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**Question 5.** Answer all parts of the following: [5 × 2 = 10]

- (a) Explain why pyridine undergoes electrophilic substitution at position 3, whereas pyridine 1-oxide reacts at position 4.
- (b) Write down the resonating structures of naphthalene and indicate the most stable resonance structure with an explanation.
- (c) Fructose has a keto functional group, but it reduces Tollens' and Fehling's reagents. Explain why.
- (d) Give the preparation of Indigotin from anthranilic acid.
- (e) Compare furan and pyrrole towards electrophilic substitution reactions.

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**Question 6.**

- (a) Write down the structure of the product of each of the following reactions: [2 × 3 = 6]

- (i) Naphthalene  $\xrightarrow{\text{alkaline KMnO}_4}$  ?
- (ii) Indigotin  $\xrightarrow{\text{dil. NaOH, Zn}}$  ?
- (iii) Pyrrole  $\xrightarrow{\text{CH}_3\text{MgBr, then CH}_3\text{I}}$  ?

- (b) Complete the following reactions, showing the mechanisms: [6]

- (i) Pyridine  $\xrightarrow{\text{NaNH}_2, \text{toluene}}$  ? (Chichibabin reaction)
- (ii) Furan  $\xrightarrow{\text{CH}_3\text{COCl, SnCl}_4}$  ?

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**Question 7.**

- (a) Write down the structures of the products of each of the following reactions: [2 × 3 = 6]

- (i)  $\beta$ -D-Glucose  $\xrightarrow{\text{dry HCl, MeOH}}$  ?
- (ii) Quinoline  $\xrightarrow{\text{HNO}_3, \text{H}_2\text{SO}_4}$  ?
- (iii) Anthracene  $\xrightarrow{\text{Na, EtOH}}$  ?

- (b) Discuss Skraup quinoline synthesis with a detailed mechanism. [6]

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**Question 8.**

(a) How will you carry out the following conversions? Write stepwise mechanisms: [6]

(i) D-Glucose  $\xrightarrow{\text{i. Br}_2/\text{H}_2\text{O}, \text{ii. Fe}^{3+}, \text{H}_2\text{O}_2}$ ? (Ruff degradation)

(ii) D-Arabinose  $\xrightarrow{\text{i. HCN, ii. H}_3\text{O}^+, \text{iii. Na-Hg}}$ ? (Kiliani-Fischer synthesis)

(b) Discuss the structures and aromatic characters of pyrrole, furan, and thiophene. [6]